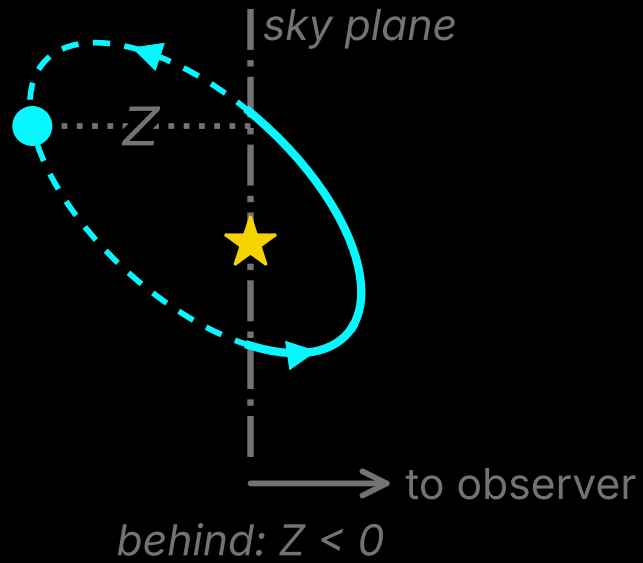


# Why is the planet half lit where it crosses the sky plane?

behind the sky plane

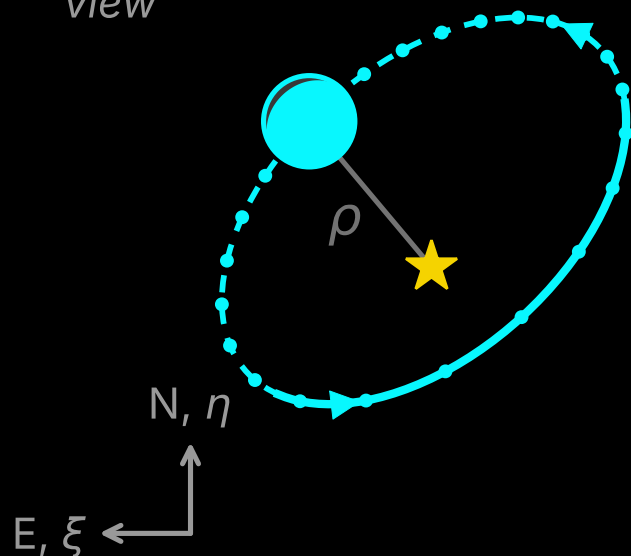
side view  
from the east



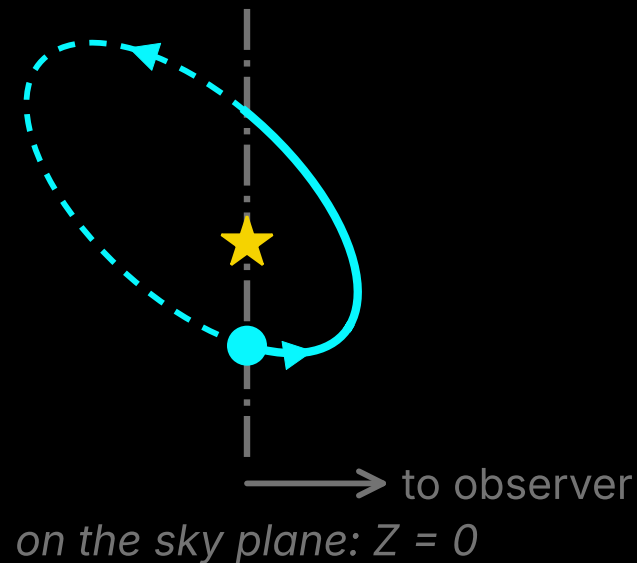
$\alpha = 35$  deg

$\Phi$  (Lambert phase) = 0.84

observer's  
view

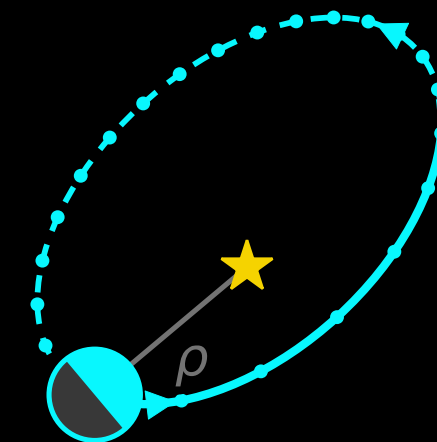


on the sky plane (ascending node)



$\alpha = 90$  deg

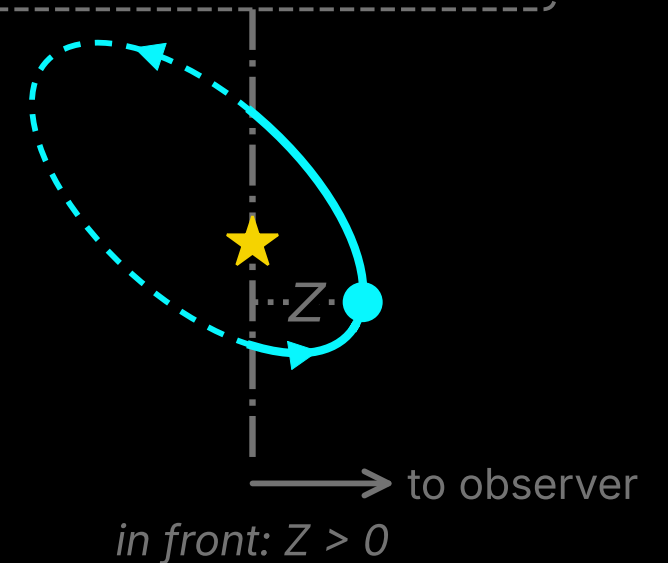
$\Phi = 0.32$



dots:  
24 equal time steps per period

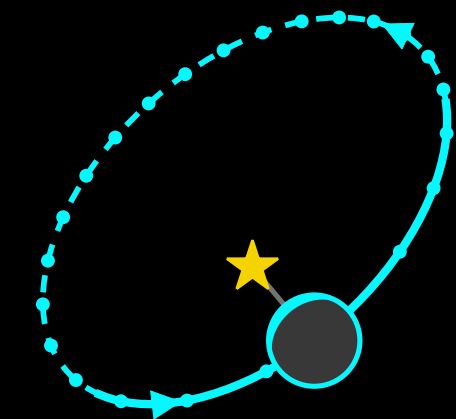
in front of the sky plane

proposed observer profile (pending)  
orbit to scale



$\alpha = 145$  deg

$\Phi = 0.02$



planet disk enlarged;  
colored = lit, gray = unlit